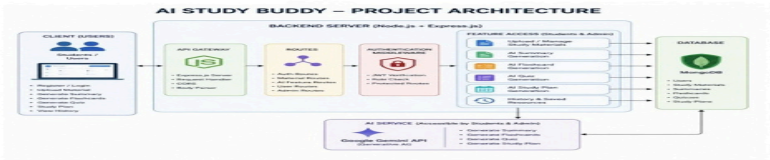
LEARNMATE – AI STUDYBUDDY

1. Brainstorming& Ideation Phase

Identifying the student study problems and shaping the AI StudyBuddy solution.

Date	21-09-26
Team ID	SWTID-2026-5714
Project Name	LearnMate - AI StudyBuddy
Maximum marks	4 Marks

Epic 1: Project Architecture



The **AI StudyBuddy** backend follows a modular RESTful architecture designed to provide secure, scalable, and AI-powered learning services. The system processes client requests through Express.js, validates users using JWT authentication, interacts with the MongoDB database for data storage, and communicates with the Google Gemini AI API to generate intelligent study resources.

The architecture separates authentication, business logic, AI services, and database operations into independent modules, making the application easier to maintain, extend, and debug.

Components Definition

- **Client Application:** The frontend (React.js) allows users to register, log in, upload study material, and access AI-powered features such as summaries, quizzes, flashcards, and study plans through REST APIs.
- **Express Server:** The Express.js server receives HTTP requests, handles routing, processes middleware, and sends appropriate responses to the client.
- **Authentication Middleware:** JWT-based middleware verifies user identity, protects private routes, and ensures that only authenticated users can access AI services and study resources.

AI-powered learning assistance and smart study support platform

Problem Background

Key Points

- Study materials are scattered across lecture notes, PDFs, textbooks, online articles and handwritten notes.
- Manual preparation of notes, flashcards and revision questions consumes significant study time.



Reference visual from the supplied LearnMate project document.

- Students need personalized study plans and a centralized place for AI-generated learning resources.

Brainstormed Solution Ideas

Key Points

- AI-generated concise summaries for faster revision.
- AI-generated flashcards for active recall learning.
- Multiple-choice quizzes for self-assessment.
- Personalized study plans based on learning goals and available study time.
- Secure user authentication and storage of generated learning resources.

Idea Direction

Key Points

- Use Google Gemini AI to dynamically generate educational content from user-provided study material.
- Use a modular backend so authentication, business logic, database operations and AI processing remain separated.